

Kelvin Olonade

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Summary

Electrical and Electronics Engineering undergraduate and open-source silicon designer working across analog, mixed-signal, and digital integrated circuit design. Two digital ASICs signed off through synthesis, place and route, DRC, LVS, and GDS generation on the SkyWater 130 nm open-source flow, with a third design, a RISC-V system-on-chip, being ported to SKY130 for the next shuttle. Currently designing a full-custom ratiometric temperature-sensor front end in IHP SG13CMOS5L 130 nm, where diagnosing and re-architecting a nonlinear input stage improved post-calibration accuracy 26 times; a charge pump and passive loop filter for a GF180MCU ring-oscillator phase-locked loop; and a dynamic-latch comparator with its layout for a 10-bit SAR ADC. Fluent in ngspice, Xschem, KLayout, Magic, Netgen, OpenLane, OpenROAD, Yosys, Cocotb, and Python-driven simulation automation.

Education

University of Lagos | Bachelor of Science, Electrical and Electronics Engineering | Lagos, Nigeria | Oct 2023 - Oct 2028 (expected)

- CGPA: 4.08 / 5.00.
- Technical focus: analog and mixed-signal IC design, digital ASIC design, RTL verification, RISC-V systems, semiconductor reliability, and open-source silicon.

Research Appointments

Taiwan Experience Education Program (TEEP) 2026 | Feng Chia University | Taichung, Taiwan | Selected Aug 2026

- Invited research intern in the Reliable Integrated Circuit Development Laboratory, Department of Electronics (Prof. Ching-Hwa Cheng), on machine-learning-assisted FinFET layout.
- Research period scheduled Sep 2026 to Dec 2026. Offer accepted; start date pending visa processing.

Technical Skills

Analog and Mixed-Signal Design: Full-custom transistor-level design, relaxation oscillators, Schmitt triggers, voltage-to-current converters, current mirrors and cascode biasing, charge pumps, passive loop filters, ring-oscillator PLLs, dynamic-latch comparators, SAR ADC architecture, bandgap and reference concepts, behavioral analog modelling, PVT-corner analysis, Monte Carlo analysis, noise, offset, metastability, settling, and linearity analysis

Analog EDA: ngspice, Xschem, KLayout, Magic, Netgen, PDK BSIM device models, parasitic extraction, post-layout simulation, DRC, LVS

Hardware Description Languages: Verilog, SystemVerilog, synthesizable RTL, parameterized modules, finite-state machines, RISC-V assembly

Digital Design: RTL microarchitecture, datapaths, control logic, AHB-Lite bus design and arbitration, QSPI memory controllers, DSP accelerators, FIR filters, UART, SPI, GPIO, timers, interrupt controllers, register and memory interfaces

Verification: Self-checking testbenches, Cocotb, constrained-random verification, functional coverage, code coverage, toggle coverage, assertions, edge-case analysis, Python golden models, gate-level simulation, SDF-annotated timing simulation

Physical Design and ASIC Flow: OpenLane, OpenROAD, Yosys, standard-cell synthesis, static timing analysis, floorplanning, macro hardening, place and route, clock and timing closure, device matching, common-centroid placement, guard rings, supply isolation, noise-aware routing, GDS generation

Semiconductor Processes: IHP SG13G2 and SG13CMOS5L 130 nm, SkyWater SKY130, GlobalFoundries GF180MCU, Tiny Tapeout shuttle flows, open-source process design kits

Simulation Automation: Python post-processing and regression analysis, NumPy, Matplotlib, bash simulation harnesses, parallel netlist generation and sweep execution, Git-based design version control

Embedded Systems: ARM Cortex-M, STM32, ESP32, Raspberry Pi Pico W, Raspberry Pi 4B and 5, PIC16F, MCU firmware, bare-metal startup code and linker scripts, sensor signal chains, EEPROM, UART, SPI, edge AI accelerators (Hailo-8)

Programming and Tools: C, C++, Python, MATLAB, Simulink, Git, GitHub, Linux, Ubuntu, Red Hat Enterprise Linux, shell-based development workflows, technical documentation

Analog and Mixed-Signal IC Design Projects

ChronoSense-1: Self-Calibrating NTC Temperature-Sensor Interface

IHP SG13CMOS5L 130 nm | Independent full-custom analog design, sole designer | HeiChips 2026 shuttle tapeout | Aug - Sep 2026

- Designing a time-domain sensor front end that converts a 10 kilo-ohm NTC thermistor ($B = 3950$ K) into a digital count using two matched relaxation oscillators, one driven by the sensor and one by a precision reference, so that supply voltage, die temperature, and common-mode drift cancel ratiometrically in the count ratio. Target: 0.1 degrees C accuracy over 0 to 70 degrees C in a 0.04 square millimeter shuttle slot.
- Diagnosed a nonlinearity in the first-pass PMOS-diode input stage using a DC sweep: sensor pad voltage drooped from 0.49 V to 0.24 V across the sensor range, making the resistance-to-period conversion nonlinear and leaving 24,700 ppm of full-scale residual, about 2.5 degrees C, after standard two-point calibration.
- Re-architected the input stage as a fixed-voltage voltage-to-current converter, cutting post-calibration residual 26 times to 934 ppm, equivalent to 0.043 degrees C, and closing the error budget with 3.7 times margin against the 0.1 degrees C target.
- Built a phase-structured ngspice validation framework (Schmitt-trigger characterization, architecture screen, PVT sweep) on IHP PDK BSIM models, driven by a bash harness that generates netlists and executes four simulations in parallel.
- Wrote the Python analysis pipeline that fits the temperature-resistance transfer, extracts post-calibration residual in parts per million of full scale, and converts residual to equivalent degrees C through the NTC sensitivity law.
- Screened 3 candidate front-end topologies across 6 sensor values and 27 PVT corners, 162 simulated conditions in total, to select the final architecture on evidence rather than intuition.
- Reduced the final analog core to 14 transistors and 2 capacitors to fit the shared-die area budget.
- Working to a September 2026 HeiChips submission deadline: transistor-level validation closing out, with layout and the digital back end following on the same schedule. Schematic entry in Xschem, floorplanning in KLayout.

Ring-Oscillator PLL: Charge Pump and Passive Loop Filter

GF180MCU 180 nm | IEEE SSCS PICO Chipathon 2026, team project | 2026 - Present

- Responsible for the charge pump and passive loop filter in a team-designed programmable ring-oscillator phase-locked loop on the GlobalFoundries 180 nm MCU open process.
- Designed the charge-pump architecture around matched source and sink current paths, cascode current-mirror biasing, and programmable current control, sizing devices from an output-voltage operating-range analysis.
- Targeted under 2 percent source-to-sink current mismatch across the usable charge-pump output swing while analyzing charge injection, clock feedthrough, static phase offset, and current-source headroom.
- Derived the passive loop filter directly from PLL system parameters (charge-pump current, oscillator gain, division ratio, target loop bandwidth, damping factor) to a phase-margin target of at least 55 degrees, then evaluated loop stability, bandwidth, reference-spur suppression, and sensitivity to component variation.
- Applied robust-layout practice to the block: device matching, symmetry, common-centroid placement where applicable, guard rings, supply isolation, and noise-aware routing.
- The team did not reach the 2026 shuttle deadline. Charge pump and loop filter are complete at schematic level and the design is being carried forward to a later tapeout opportunity.

10-Bit 1-MS/s SAR ADC: Dynamic-Latch Comparator and Layout

IHP SG13CMOS5L 130 nm | Chipalooza Challenge 2026, team project | 2026 - Present

- Own the dynamic-latch comparator in a team-built 10-bit differential-input successive-approximation ADC intended as reusable mixed-signal IP for sensor interfaces, monitoring circuits, and SoC integration: schematic design, characterization, and layout.
- Characterize decision polarity, DC input-referred offset, transition point, regeneration time, decision-time budget, noise behavior, metastability probability, and PVT robustness.
- Size the comparator against a 15 MHz per-bit decision budget so that all ten bit trials plus sampling close inside the 1 MS/s conversion window with timing margin across corners.
- Draw the comparator layout for differential-pair matching, common-centroid placement, guard rings, and supply isolation, and close DRC and LVS in the open-source IHP flow.
- Review neighboring blocks (sampling network, differential CDAC, SAR control logic, voltage references, level shifters) during integration and correct issues found.

Digital ASIC and Silicon Projects

RV32E RISC-V SoC with 16-Tap FIR Accelerator and External QSPI PSRAM

Designed on GF180MCU, porting to SkyWater SKY130 | Tiny Tapeout, 2 tiles | Targeting the next SKY130 shuttle

- Designing a minimal RV32E RISC-V system-on-chip with a custom 16-tap finite impulse response accelerator, UART, GPIO, timer, interrupt controller, and an AHB-Lite crossbar, hardened for a two-tile Tiny Tapeout slot.
- Moved program and data memory off-chip after the area budget showed on-chip memory dominating the floorplan (roughly 48,000 of 74,000 square micrometers), replacing 3 KB of on-chip ROM and SRAM with 8 MB of external QSPI PSRAM: about 2,700 times the addressable memory, in a fraction of the die area, and a Harvard machine converted to a flat von Neumann map.
- Wrote a QSPI PSRAM controller (AHB-Lite shim slave plus mode-0 QSPI master at 12.5 MHz SCK) with a two-word burst read and a one-deep prefetch line, cutting the average cost of sequential instruction fetch from roughly 60 to roughly 31 cycles per word. Writes are sized from the bus transfer size and invalidate the prefetch line.
- Designed a roughly 150-gate hardware boot loader that holds the CPU in reset, streams a length-prefixed flat firmware image into PSRAM over a four-phase host handshake (about 4 ms for a 4 KB image), then releases the CPU to fetch from address zero. This removes mask-ROM respin risk entirely and collapses startup code to a stack-pointer write and a jump, with no data copy or BSS zero loop.
- Caught a fatal bus-handshake defect in design review: memory ready was asserted combinationally in the request cycle while every slave presents read data one cycle later, so the core would have consumed data from the previous transaction on every access and executed garbage from the first instruction fetch. Replaced it with a one-cycle-delayed registered-response handshake.
- Caught a second boot-blocking defect where the interrupt controller reset with all sources unmasked, vectoring the CPU into an unwritten handler address on the first accelerator completion, and re-specified the reset default.
- Implemented the open-source ASIC flow from synthesizable RTL through Yosys synthesis, OpenLane macro hardening, gate-level simulation, and SDF-annotated timing verification.
- Validated FIR mathematical accuracy against a Python golden model within plus or minus 1 LSB, and measured at least 4 times acceleration over a software FIR implementation on the same core.
- Built bare-metal firmware in C and RISC-V assembly, including custom startup code, memory initialization, and a linker script.
- Porting the design from GF180MCU to SkyWater SKY130 for the next Tiny Tapeout shuttle: re-running synthesis and macro hardening against SKY130 standard cells, re-closing timing at the new library delays, and re-checking the two-tile area budget, where the 16-by-16 multiplier in the FIR datapath is the dominant risk.

Hardware EEPROM Wear-Leveling and Address-Scrambling Controller

SKY130 | Tiny Tapeout TTSKY26B shuttle | Taped out | github.com/OlonadeKelvin/EEPROM_Wear

- Designed and taped out a compact hardware wear-leveling controller based on the Start-Gap algorithm to extend EEPROM and flash lifetime under repeated write cycles, motivated by a field failure investigated at Balancee Tech Solutions.
- Added a two-round Feistel address scrambler with an LFSR-derived key to defeat predictable physical-address targeting and spread writes across memory blocks.
- Implemented 8-bit per-block wear counters, bad-block retirement, logical-to-physical address translation, and in-band telemetry for total writes and maximum-to-minimum wear skew.
- Developed Cocotb functional verification covering address translation, counter behavior, bad-block handling, telemetry, reset behavior, and repeated-write stress cases.
- Completed synthesis, place and route, GDS generation, and simulation sign-off through the SkyWater 130 nm open-source ASIC flow.

Autonomous VGA Algorithmic Synthesizer ASIC

SKY130 | Tiny Tapeout TTSKY26A shuttle | Taped out | github.com/OlonadeKelvin/DemosceneTTSKY

- Designed a tapeout-ready digital ASIC that autonomously generates 640-by-480 VGA graphics from on-chip fractal and linear-feedback shift-register engines, with no firmware and no external controller.
- Wrote synthesizable Verilog and SystemVerilog RTL for VGA timing, fractal generation, pseudo-random visual effects, pixel composition, and output control.
- Hand-optimized the architecture into an approximately 160-by-100 micrometer Tiny Tapeout tile and pipelined the pixel datapath to close timing at the 25.175 MHz pixel clock.
- Balanced critical paths against area using synthesis and static-timing feedback from the OpenLane flow.
- Developed a self-checking SystemVerilog testbench with functional coverage for horizontal and vertical synchronization, fractal correctness, timing behavior, and entropy quality.

Additional Engineering Projects

Synthesizable UART Intellectual Property Core

Verilog | Digital design | github.com/OlonadeKelvin/Synthesizable-UART-Transmitter-Receiver-IP-Core

- Designed a synthesizable UART core with transmit, receive, baud-rate control, finite-state-machine sequencing, and simulation-based verification.

Firmware-Free Flood Monitoring System

PIC16F | Configurable logic blocks

- Built a flood-monitoring proof of concept using PIC16F configurable logic blocks and core-independent peripherals to perform control functions with minimal firmware intervention.

Multicore Processor Thermal Simulator

MATLAB and Python | Engineering simulation | github.com/OlonadeKelvin/Multicore-Processor-Die

- Developed a computational model for temperature behavior and thermal interaction in multicore processor systems.

Experience

Syprion Labs | Founder and Lead Engineer | Lagos, Nigeria | Jul 2026 - Present

- Founded and lead an engineering studio focused on semiconductor design, embedded systems, hardware development, engineering simulation, and practical technology education.
- Define technical direction across digital hardware, embedded controllers, open-source silicon, engineering software, and hardware-software integration.
- Lead project planning, system architecture, design reviews, technical documentation, prototyping, debugging, and delivery coordination.
- Build reusable engineering workflows and internal knowledge from prior freelance work in embedded systems, simulation, and research prototyping.
- Support engineers and students in translating technical concepts into verifiable hardware, firmware, simulation models, and documented prototypes.

Balancee Tech Solutions | Embedded Systems Engineer | Ajah, Lagos, Nigeria | May 2026 - Present

- Led end-to-end development of a custom fuel-pump embedded controller: schematic capture, microcontroller firmware, sensor signal-chain design, system integration, and field-test verification.
- Integrated hardware and software into an embedded control system matched to application requirements and field operating conditions.
- Investigated persistent EEPROM endurance failures caused by frequent field calibration writes, isolated the write-concentration mechanism, and carried the failure analysis into a hardware wear-leveling controller later taped out on SkyWater 130 nm.
- Maintain proprietary project documentation, schematics, design records, and implementation details under confidentiality.

AVZDAX | Embedded Systems Intern | Victoria Island, Lagos, Nigeria | Dec 2025 - May 2026

- Delivered an edge computer-vision pipeline on Raspberry Pi 5 and Raspberry Pi 4B using a Hailo-8 AI accelerator running a custom-trained proprietary model.
- Profiled camera capture, preprocessing, memory movement, and accelerator hand-off to locate and remove processing bottlenecks.
- Achieved stable throughput above the 30 FPS real-time target on the integrated Raspberry Pi 5 and Hailo-8 platform.
- Supported system debugging, integration testing, performance characterization, hardware bill-of-materials review, and technical documentation.

Freelance | Embedded Systems and Simulation Engineer | Lagos, Nigeria | Mar 2025 - Jul 2026

- Developed simulation models and computational algorithms for renewable energy, control systems, photonics, communication networks, and related engineering research.
- Designed microcontroller-based hardware prototypes for undergraduate and postgraduate research projects.
- Integrated embedded firmware with custom electronic hardware to build proof-of-concept systems for thesis, capstone, and research applications.
- Validated simulation models through laboratory testing, hardware implementation, debugging, and performance analysis.
- Prepared technical documentation, simulation reports, experimental results, and demonstrations for thesis evaluations and technical presentations.
- Collaborated with students and academic supervisors to translate research concepts and mathematical models into functional hardware and software.

Leadership and Professional Activities

IEEE Solid-State Circuits Society Student Chapter, University of Lagos | Founding Chair | Lagos, Nigeria | 2025 - Present

- Established the first IEEE Solid-State Circuits Society student chapter at the University of Lagos.
- Organize RTL, ASIC, semiconductor, and integrated-circuit design workshops for engineering students.
- Build partnerships, learning programs, and peer-development opportunities to strengthen the pipeline of semiconductor engineers in Nigeria.
- Support students entering digital design, verification, analog and mixed-signal circuits, physical design, open-source EDA, and tapeout programs.

IEEE Student Societies, University of Lagos | Vice Chair | Lagos, Nigeria | 2025 - Present

- Vice-chair roles across the Industrial Electronics, Industry Applications, Circuits and Systems, Computer, Signal Processing, and Power and Energy Societies.

Arm Developer Program | Contributor | Remote | 2025 - Present

- Apply Arm developer documentation, architecture references, and peripheral guidance to practical Cortex-M firmware and hardware-integration problems.

Technical Education and Mentoring | Instructor and Community Contributor | Lagos, Nigeria | 2024 - Present

- Teach and support undergraduate and postgraduate students working on embedded systems, engineering simulation, digital design, and research prototypes.
- Help organize technical hackathons, workshops, project reviews, and peer-learning activities.

Publications

A Guide to Embedded Systems Engineering. Technical educational publication introducing embedded-system concepts, development workflows, hardware-software integration, and practical engineering considerations.

Certifications and Training

- Zero to ASIC Course.
- ARM Cortex-M Processors Overview.
- Mastering Pointers in C: A Course on Efficient Programming.
- C for Everyone: Programming Fundamentals.
- C for Everyone: Structured Programming, with Honors.
- MATLAB and Simulink Training.
- Designathon Certificate of Participation.

Languages

English: native or bilingual proficiency. Yoruba: professional working proficiency. German: elementary proficiency. Chinese: elementary proficiency.

Interests

Semiconductor technology, analog and mixed-signal design, open-source silicon, technical writing, engineering education, research planning, hardware experimentation, and racing games.